

Neural pattern similarity predicts long-term fear memory

Renée M Visser^{1,2}, H Steven Scholte^{2,3}, Tinka Beemsterboer³ & Merel Kindt^{1,2}

Although certain changes in the brain may reflect fear learning, there are no known markers that indicate whether an aversive experience will develop into fear memory. We examined the moment-to-moment dynamics of human fear learning by applying multi-voxel pattern analysis to single-trial blood oxygen level-dependent magnetic resonance imaging data. We found that the long-term behavioral expression of fear memory could be predicted from neural patterns at the time of learning.

Despite everything that we know about the processes that can weaken or strengthen memory for fearful events, our understanding of how these events are represented, processed and eventually engraved into the neural architecture of the brain remains poor. An inherent restriction of memory research, including fear conditioning, is that we can only infer fear memory from the degree to which behavior during learning (for example, freezing in rats, physiological responding in humans) overlaps with behavior at a later retention test. Although the expression of fear during learning is certainly related to long-term memory, much of what we learn does not eventually transform into long-term memory. Currently, no signature exists that permits a read-out of subsequent consolidation at the time of learning¹.

The application of multi-voxel pattern analysis (MVPA), a technique for analyzing distributed patterns of blood oxygen level-dependent magnetic resonance imaging (BOLD MRI) data², may be especially promising for fear memory research, given that there is naturally more information in patterns than in mean activation^{2,3}. Using this technique, it has been shown that the neural representations of stimuli change as a function of fear conditioning^{4–6}. Although it is tempting to interpret these changes as a marker for fear memory, they were not examined in relation to a later, independent memory test^{4–6}. The question is whether patterns of activation also inform about upcoming consolidation processes, indicating that, at least in fear memory research, MVPA is more than a cross-validation technique for assessing transient learning-dependent changes. Although neural pattern similarity during encoding can successfully predict relatively short-term (1–6 h) declarative memory performance⁷, this effect could be explained by enhanced conscious processing of the subsequently remembered items^{7,8}. Central to fear memory, however, is not the conscious processing of threat and the factual recollection of events, but the automatic emotional expression of fear.

We tested whether the formation of long-term procedural fear memory can be assessed by neural pattern similarity during learning. To this end, we combined functional MRI (fMRI) with a behavioral measure of associative fear learning (that is, pupil dilation responses) during differential fear conditioning and a memory test (**Supplementary Fig. 1a**). We repeatedly presented six stimuli, derived from two distinct categories (faces and houses), as the conditioned stimuli (CS). One face and one house co-terminated with a mild electric stimulus (unconditioned stimulus) in half of the trials (CS⁺_{neg}; **Supplementary Fig. 1b**). Another face and another house co-terminated with a neutral sound (neutral associative stimulus) in half of the trials to control for non-aversive associative learning (CS⁺_{neut}). The third face and house were never reinforced (CS⁻). For the fMRI analyses, we only analyzed CS⁺_{neg} and CS⁺_{neut} trials in which no shock or sound was presented, and corresponding CS⁻ trials (together referred to as target trials; **Supplementary Fig. 1c**).

We employed trial-by-trial similarity analysis^{6,9} to examine the formation, activation and extinction of fear associations, and to assess

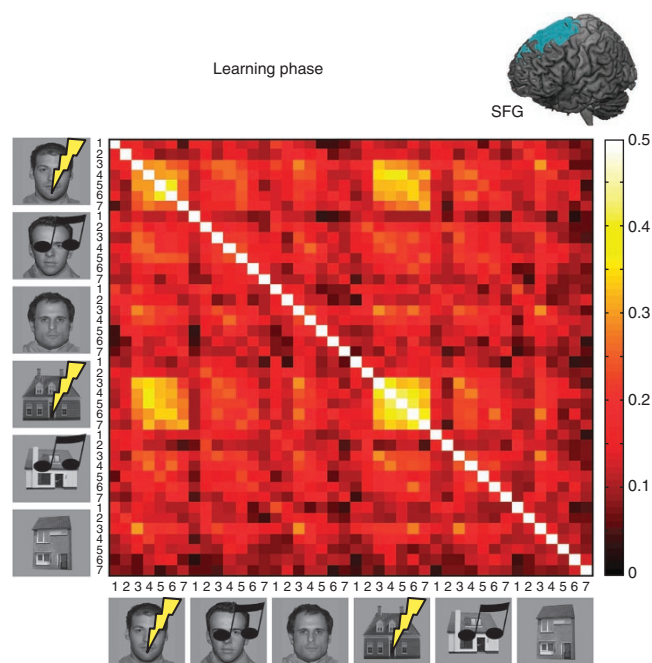
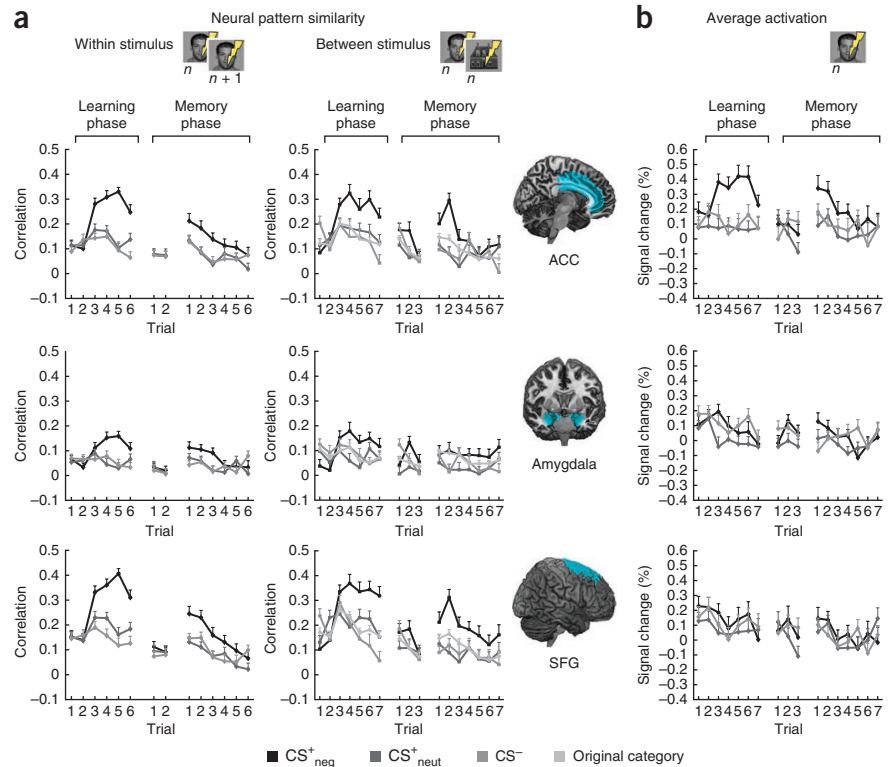


Figure 1 A 42 × 42 correlation matrix containing within-stimulus and between-stimulus correlations of BOLD MRI patterns in the SFG during the learning phase ($n = 38$). The off-diagonal represents correlations between two consecutive target trials (within-stimulus correlations). Between-stimulus correlations include correlations between stimuli from the same original category (faces and houses) and between stimuli that share a particular outcome (shock versus sound versus no consequence).

¹Department of Clinical Psychology, University of Amsterdam, Amsterdam, The Netherlands. ²Cognitive Science Center Amsterdam, University of Amsterdam, Amsterdam, The Netherlands. ³Department of Brain and Cognition, University of Amsterdam, Amsterdam, The Netherlands. Correspondence should be addressed to M.K. (m.kindt@uva.nl).

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Figure 2 Neural pattern similarity versus average activation during different experimental phases. **(a,b)** Within-stimulus pattern similarity (averaged over faces and houses), between-stimulus pattern similarity **(a)** and average single-trial activation (averaged over faces and houses) **(b)** in the ACC, amygdala and SFG ($n = 38$) revealed the formation, re-expression and extinction of fear associations. The high sensitivity of similarity analysis compared with univariate analysis revealed the involvement of brain areas outside the traditional fear circuit (for example, SFG). Error bars represent s.e.m.



whether pattern similarity predicts the long-term expression of fear memory. We modeled trials as separate events. For each of the 38 participants, we created a vector containing the spatial pattern of BOLD MRI signal related to one stimulus presentation for six anatomically defined regions of interest (ROIs): the anterior cingulate cortex (ACC), the insula, the amygdala, the hippocampus, the ventromedial prefrontal cortex (vmPFC) and the superior frontal gyrus (SFG). Next, we correlated each vector with all other vectors related to the other stimulus presentations, resulting in a similarity matrix of 42×42 target trials (Fig. 1). Correlations of interest included the correlations between two consecutive presentations of the same stimulus (within stimulus) and between pairs of different stimuli (between stimulus), including stimuli that belonged to the same category (original categories, faces and houses) and stimuli that shared a specific outcome (categories based on learned association, shock versus sound versus no consequence) (Supplementary Fig. 2). Between-stimulus pattern similarity is characterized by a focus on a common outcome rather than a focus on single predictors and putatively reflects a type of higher order fear learning⁶.

Consistent with previous work⁶, the application of trial-by-trial similarity analysis revealed clear learning curves that indexed the formation of associative fear (within stimulus: trial[6] $\times$ stimulus[3], between stimulus: trial[7] $\times$ stimulus[4]; all $P \leq 0.004$, $P\eta^2$ ranging from 0.06–0.22; Fig. 2a, Supplementary Fig. 3a and Supplementary Table 1). Participants returned for the memory phase 2–6 weeks later (mean = 22.18 ± 6.4 d). Without electrodes attached, exposure to previously learned CS^+_{neg} stimuli did not elicit differential pattern similarity. With electrodes attached, differential pattern similarity reappeared and, as the aversive outcome was no longer delivered, eventually extinguished (Supplementary Table 2). To compare our approach with standard univariate analysis, we examined single-trial activation, averaged over all voxels in a ROI. This yielded similar learning and extinction curves as were obtained with similarity analysis in ACC and insula (Fig. 2b), although interaction effects did not reach FDR-corrected significance (all $P \geq 0.033$; Supplementary Table 1).

In parallel with the results from the similarity analysis, successful fear conditioning was also evident from increased pupil dilation in response to CS^+_{neg} stimuli compared with CS^+_{neut} and CS^- stimuli over the course of conditioning (trial[13] $\times$ stimulus[3], $F_{24,816} = 9.15$, $P < 0.0005$, $P\eta^2 = 0.21$). Without electrodes attached, the presentation of previously learned CS^+_{neg} stimuli did not elicit enhanced pupil responses. With electrodes attached, differential pupil responses were again present (main effect of stimulus, $P < 0.0005$, $P\eta^2 = 0.43$)

and eventually extinguished (trial[13] $\times$ stimulus[3], $F_{24,816} = 2.49$, $P < 0.0005$, $P\eta^2 = 0.07$; Supplementary Fig. 4). These findings confirm the validity of the conditioned pupil dilation response as a behavioral measure of fear¹⁰.

After the first three trials of the extinction phase, differential pupil responses rapidly diminished (second versus third extinction trial, $F_{1,37} = 0.14$, $P = 0.713$; third versus fourth extinction trial, $F_{1,37} = 8.10$, $P = 0.007$), indicating that extinction learning started after the third trial. To predict the long-term expression of fear memory, we classified the 38 participants according to their behavioral expression of fear memory (Fig. 3a), averaged over these first three retention trials before extinction learning became apparent. We assigned participants to the retention group ($n = 22$) if they showed a stronger pupil response to the CS^+_{neg} face stimulus as well as to the CS^+_{neg} house stimulus, as compared with any of the four control stimuli (CS^+_{neut} and CS^-). The remaining participants were assigned to the no retention group ($n = 16$). Groups did not differ on subjective^{11,12} and procedural variables (Supplementary Table 3 and Supplementary Fig. 5).

We assessed the predictive value of fear learning curves derived from neural pattern similarity by comparing groups on mean conditioning indices (Online Methods), which expressed the relative increase of the CS^+_{neg} responses over the course of learning. For within-stimulus pattern similarity, no substantial differences were found (Supplementary Table 4 and Supplementary Fig. 6). For between-stimulus pattern similarity, we found more stimulus differentiation during initial fear learning in the retention group than in the no retention group in the ACC ($F_{1,36} = 6.53$, $P = 0.015$, $P\eta^2 = 0.15$), insula ($F_{1,36} = 5.10$, $P = 0.030$, $P\eta^2 = 0.12$), amygdala ($F_{1,36} = 4.96$, $P = 0.032$, $P\eta^2 = 0.12$), hippocampus ($F_{1,36} = 5.16$, $P = 0.029$, $P\eta^2 = 0.13$) and vmPFC ($F_{1,36} = 8.47$, $P = 0.006$, $P\eta^2 = 0.19$; Fig. 3b), and observed a trend in the SFG ($P = 0.079$) (Supplementary Fig. 7 and Supplementary Table 5). Notably, the retention and no retention groups did not differ

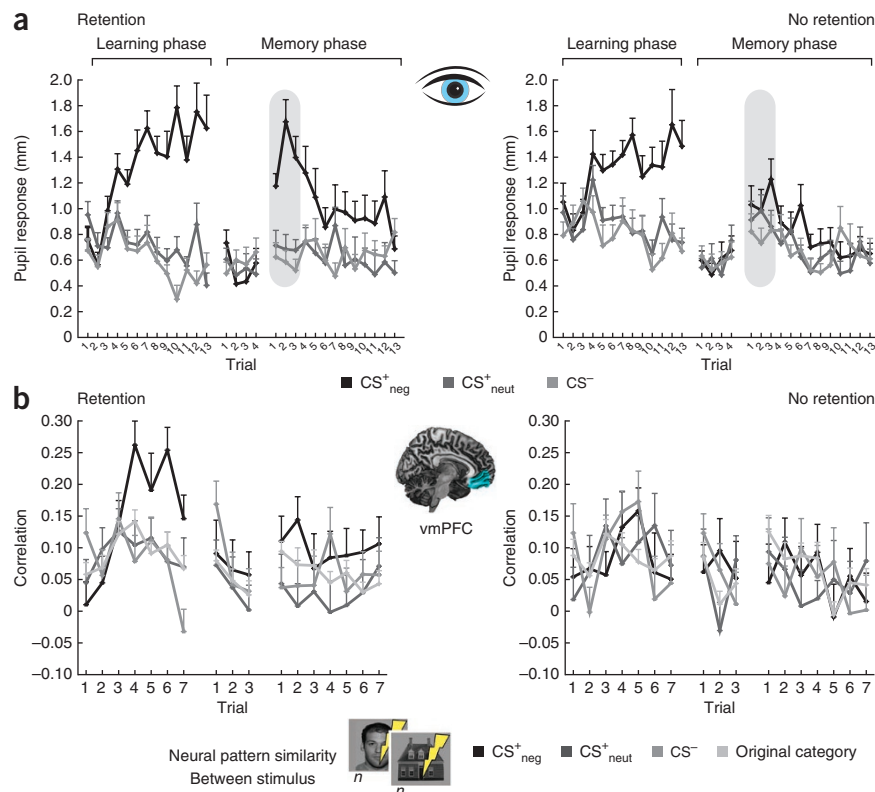
Figure 3 Pupil dilation and neural pattern similarity split by long-term procedural fear memory. (**a,b**) Pupil dilation responses ($n = 35$) showed differential learning and extinction of fear. On the basis of differential pupil responses during the first three trials from the extinction phase, participants were assigned to the retention group ($n = 19$ for pupil data, $n = 22$ for MRI data, **a**) or the no retention group ($n = 16$, **b**). Although differential pupil responses during learning did not predict subsequent pupil responses, differential pattern similarity in several brain regions, including the vmPFC (**b**), did. Error bars represent s.e.m.

in pupil responses at the time of fear learning ($F_{1,33} = 2.19$, $P = 0.148$; **Fig. 3a**).

Follow-up tests for both groups separately (**Supplementary Table 6**) revealed strong fear learning, as indexed by differential pattern similarity, in the retention group in all areas (all $P \leq 0.018$; **Fig. 3b** and **Supplementary Fig. 7**). The no retention group showed a different pattern. For this group, no fear learning was observed in the amygdala, hippocampus (**Supplementary Fig. 7**) and vmPFC (**Fig. 3b**), but some fear learning was observed in the ACC, the insula and the SFG (**Supplementary Fig. 7**). A continuous assessment of the predictive value of pattern similarity revealed a linear relationship between differential between-stimulus pattern similarity in the ACC, insula, vmPFC and SFG during fear learning and subsequent differential pupil dilation during the retention trials (**Supplementary Table 7**). Again, differential pupil responses during fear learning did not predict differential pupil responses at test ($P = 0.464$).

Crucially, we found that average activation did not predict the later expression of fear memory in any of the ROIs (**Supplementary Fig. 8** and **Supplementary Tables 7** and **8**) or anywhere else in the brain (**Supplementary Table 9**). We repeated the ROI analyses while only considering voxels that were responsive to the task². This enhanced learning effects for both pattern similarity and mean activation (**Supplementary Table 10**), but again revealed that the prediction of procedural fear memory was restricted to between-stimulus pattern similarity (**Supplementary Tables 4**, **5** and **8**). Although within-stimulus pattern similarity did not clearly predict procedural fear memory, we replicated other findings on declarative memory⁷ by showing that, in the hippocampus, within-stimulus pattern similarity was higher for subsequently remembered stimuli than for subsequently forgotten stimuli ($F_{1,15} = 5.28$, $P = 0.036$, $P\eta^2 = 0.26$, uncorrected; **Supplementary Table 11**).

In sum, the refinement of an individual stimulus representation (within stimulus) seems to be different from the formation of a higher order fear association (between stimulus). The cortical distribution of between-stimulus and within-stimulus pattern similarity further substantiates this idea⁶. Although these two types of neural pattern similarity both reflect fear and extinction learning, they apparently contain different information about future consolidation processes, given that only between-stimulus neural pattern similarity predicts the later behavioral expression of fear memory. Higher order fear learning expressed as changes in neural patterns may offer a promising signature to examine the determinants of fear memory.



METHODS

Methods and any associated references are available in the [online version of the paper](#).

Note: Supplementary information is available in the [online version of the paper](#).

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AUTHOR CONTRIBUTIONS

R.M.V., H.S.S. and M.K. designed the research. R.M.V. and T.B. performed the research and analyzed the data. H.S.S. contributed new analytic tools. H.S.S. and M.K. supervised the project. R.M.V. and M.K. wrote the manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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ONLINE METHODS

Participants. 54 students of the University of Amsterdam participated in the initial fear learning session. 47 participants returned for the subsequent memory phase. Data were discarded from analyses if participants did not return for the second session or if fMRI data were confounded by substantial head motion (>2 mm in any direction, $n = 4$) or by excessive sleepiness, as judged on the basis of eye-tracker data combined with self-report ($n = 3$). In addition, two participants were excluded from the analysis because they had not learned the contingencies. For the fMRI analyses, the final sample included 38 participants (15 male, 36 right-handed) between 18 and 33 years of age (23.7 ± 3.8 , mean $\pm$ s.d.). The time between the two sessions ranged from 13 to 43 d (22.18 ± 6.4 , mean $\pm$ s.d.). From three participants, part of the eye-tracker data was missing (not the retention trials) and from six participants part of the associative stimulus expectancies was missing. Thus, when data from the three experimental phases are compared directly, eye-tracker data are reported for 35 participants and unconditioned stimulus and associative stimulus expectancies are reported for 32 participants. Participants earned course credit or €22 per session. All participants gave their written informed consent before participation and were naive to the purpose of the experiment. Procedures were executed in compliance with relevant laws and institutional guidelines, and were approved by the ethics committee of the University of Amsterdam.

Experimental design. The experiment consisted of two phases: a learning phase and a memory phase. The latter consisted of a baseline and an extinction phase (Supplementary Fig. 1a). For the learning phase, a differential fear conditioning procedure was used, with delay conditioning and partial reinforcement (Supplementary Fig. 1b). Three faces (NimStim set¹³) and three houses (collected from the internet and separated from their background) were repeatedly presented for 4.5 s and served as the to-be-conditioned stimuli. Stimuli were converted to grayscale and presented on a gray background.

The electrical stimulation served as aversive unconditioned stimulus and was delivered twice for 2 ms, with a delay of 300 ms, applied by a Digitimer DS7A through MRI-compatible carbon electrodes attached to the right shin bone. Prior to the experiment, the intensity of the electric stimulus was individually adapted to be aversive, but not painful (intensity range = 8–71 mA, mean $\pm$ s.d.: 34.6 ± 15.6). For the neutral associative stimulus, a sound (500 ms) was selected that is standard implemented in Windows (chimes). Subjective assessments (Supplementary Table 3) confirmed the intended valence of both the unconditioned and associative stimuli.

Inter-stimulus intervals were fixed and were long enough (19.5 s) to limit intrinsic noise correlations. The onset of each trial was triggered by the start of the acquisition of a BOLD MRI volume. The order of stimulus presentation was fixed (counterbalanced across participants) and consisted of a repeating sequence of six target trials, with filler trials of the same stimuli in between (Supplementary Fig. 1c). In total, the learning phase consisted of 78 trials: 42 target trials (seven per stimulus type) and 36 filler trials (six per stimulus type), including all CS^+_{neg} trials that co-terminated with a shock and all CS^+_{neut} trials that co-terminated with a sound. The baseline phase (without electrodes) consisted of 24 trials: 18 target trials (three per stimulus type) and six filler trials (one per stimulus type). Finally, the extinction phase (with electrodes) consisted of 78 trials: 42 target trials (seven per stimulus type) and 36 filler trials (six per stimulus type). For fMRI analyses, we constrained our analyses to target trials⁶.

Pupil dilation. Pupil dilation responses and eye movements were recorded continuously throughout MRI scanning using a remote non-ferromagnetic infrared Eyelink-1000 Long Range Mount eye tracker (SR Research). Before task onset, a nine-point calibration procedure was performed.

Data were sampled at 250 Hz. The baseline pupil diameter was the average value during the 500 ms before each conditioned stimulus onset. The pupil response to the conditioned stimulus was calculated as the peak change from baseline in a window from 0 to 4 s after picture onset, discarding any data samples that were obscured by eye blinks. Trials that suffered substantial signal loss (not related to sleepiness), affecting either the baseline or more than 2 s within 4 s of stimulus onset were eliminated (0–19% per participant, median = 0.6%) and replaced using the linear trend at point. Next, data were Fisher-transformed for each of the three experimental phases separately.

Image acquisition. Scanning was performed on a 3T Philips Achieva XT MRI scanner using a 32-channel head-coil. Functional data were acquired using gradient echo, echo planar imaging (repetition time = 2,000 ms, time to echo = 27.63 ms, flip angle = 76.1° , 39 sagittal slices with interleaved acquisition, $3.0 \times 3.0 \times 3.3$ mm voxel size, 64×64 matrix, $192 \times 192 \times 141.24$ field of view) covering the whole brain. The learning phase consisted of 946 dynamics, the baseline of 295 and the extinction phase again of 946 dynamics. Foam pads minimized head motion. A high-resolution three-dimensional T1-weighted image (repetition time = 8.124 ms, time to echo = 3.72 ms, flip angle = 8° , $1 \times 1 \times 1$ mm voxel size, $256 \times 256 \times 160$ field of view) was also collected for anatomical visualization.

Preprocessing. FEAT (FMRI Expert Analysis Tool) version 4.1.6, part of FSL (Oxford Centre for Functional MRI of the Brain (FMRIB) Software Library, <http://fsl.fmrib.ox.ac.uk/fsl/fslwiki/>) was used to analyze the fMRI data. Pre-processing steps included slice-time correction, motion correction, spatial smoothing¹⁴ using a 5 mm full-width-at-half-maximum Gaussian kernel, high-pass filtering in the temporal domain ($\sigma = 100$ s) and prewhitening¹⁵. Structural images were co-registered to the functional images and transformed to MNI standard space (Montreal Neurological Institute) using FLIRT (FMRIB's Linear Image Registration Tool, FSL). The resulting normalization parameters were applied to the functional images.

ROI selection. ROIs were selected on the basis of their role in fear learning and (extinction) memory^{16,17} and included the ACC, the insula, amygdala, hippocampus and vmPFC. We also included the SFG to illustrate that robust learning-dependent changes, as revealed by similarity analysis, can also be observed outside the salience network⁶. ROIs were obtained from the Harvard-Oxford cortical and subcortical structural atlases (Harvard Center for Morphometric Analysis). Additional analyses were performed with functional and equal-sized ROIs (Supplementary Tables 4, 5, 8 and 12).

Univariate analysis. To visualize trial-by-trial learning dependent changes in average activation, we modeled all trials as separate events and extracted for each event the average response amplitude per ROI using Matlab (version 7.11; MathWorks). For additional whole brain analyses, see Supplementary Table 9.

Trial-by-trial similarity. For the trial-by-trial representational similarity analysis, trials were also modeled as separate events. The resulting single-trial data were further analyzed in Matlab by calculating pair-wise Pearson correlations between event-related spatial patterns of activation (Z values per voxel). This resulted in a similarity matrix containing correlations among trials for each participant and for each ROI. From this matrix (Fig. 1), three different types of correlations were selected (Supplementary Fig. 2). The strength of these correlations was used as a metric of similarity. Correlations were then Fisher-transformed for each experimental phase separately. Figures, however, display raw data, as this facilitates interpretation of the results.

Statistical analyses. Fisher-transformed pupil dilation responses, within-stimulus correlations and correlations for original-related stimuli were averaged over face and house stimuli. This was done to reduce the number of comparisons and because we were not interested in the difference between faces and houses with regard to the experimental manipulation (but see Fig. 1).

Statistical comparisons of the learned associations were performed by within-subjects analysis of variance using Statistical Package for the Social Sciences (version 17, SPSS). Differential fear learning was assessed by the interaction of all trials $\times$ stimulus, but was only tested when there was also a significant main effect of stimulus type. For the baseline phase, no main effects of stimulus type were expected. Reactivated fear memory was assessed by a main effect of stimulus type during the extinction phase and, if significant, extinction was assessed by a significant interaction of all extinction trials $\times$ stimulus type.

The predictive value of the different measures was examined by assigning participants to a retention group and a no retention group, according to their pupil dilation responses on the first three extinction trials. To compare groups, we first calculated difference scores for each trial from the learning phase: $2(CS^+_{neg}) - CS^+_{neut} - CS^-$ for single-trial activation data and pupil data; $3(CS^+_{neg}) - CS^+_{neut} - CS^-$ —original category for between-stimulus correlations.

We then subtracted the average difference over the first target trials (when participants did not know the contingencies; **Supplementary Fig. 1c**) from the average difference over the later target trials (when contingencies could have been learned), yielding a conditioning index that expressed the relative increase of CS⁺_{neg} responses over the course of learning. These conditioning indices were then compared between the retention and no retention group and also used for the continuous assessment of fear memory.

Predictions were tested while correcting for multiple comparisons (six ROIs) by limiting the false discovery rate¹⁸. In the case in which the

assumption of sphericity was violated, a Greenhouse-Geisser correction was applied. All *P* values are reported two-sided.

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